International Journal of Scientific Research in Engineering and Management (IJSREM)
Volume: 10 Issue: 05 | May - 2026 SJIF Rating: 8.659 ISSN: 2582-3930

AQUIS: A Role-Based Groundwater Monitoring and Information Interface for CGWB DWLR Data

Dinesh Kumar[1] ,Srijan Anand Gupta[2], Utkarsh Srivastava[3]
[1]Assistant Professor at Bhagwan Parshuram Institute of Technology, Rohini, New Delhi, India
[2,3]Students at Bhagwan Parshuram Institute of Technology, Rohini, New Delhi, India

Email: [1]dineshkumar@bpitindia.com, [2]guptasrijan85@gmail.com, [3]utkarsh3904@gmail.com

Abstract—

India's groundwater monitoring network, operated by the Central Ground Water Board (CGWB), generates high-frequency data through thousands of Digital Water Level Recorder (DWLR) stations. However, the operational accessibility of this data remains limited for many stakeholders. Existing platforms such as India-WRIS and IN-GRES are primarily designed for technical users and provide limited support for role-specific operational workflows.

This paper presents AQUIS (Aquifer Query and User Information System), a prototype groundwater monitoring and information platform designed for citizens, district officers, and researchers. The study uses analysis of the CGWB DWLR 2023 dataset (n = 730 valid assessment units) to inform interface architecture and feature prioritisation.

The analysis showed that 85.3% of assessment units fall within the safe category, while 7.6% are over-exploited. High groundwater stress was concentrated in districts across Delhi, Bihar, Kerala, Rajasthan, and Lakshadweep. Rainfall correlation with groundwater availability was found to be negligible (r = 0.003), indicating stronger influence from extraction-related factors.

The AQUIS platform applies role-specific dashboards, groundwater stress visualisation, hotspot mapping, configurable alerts, and analytical tools suited to different user groups. Prototype evaluation through heuristic review identified six usability issues, resulting in corresponding interface refinements. The system is designed to remain compatible with existing CGWB and India-WRIS infrastructure through standard interoperability approaches.

Keywords— Groundwater Monitoring, DWLR, CGWB, Groundwater Governance, Role-Based Interface, India-WRIS

1. Introduction

India is the world's largest extractor of groundwater, withdrawing nearly 250 billion cubic metres annually, placing significant stress on aquifer systems across many regions (CGWB, 2023). The Central Ground Water Board (CGWB) operates an extensive DWLR monitoring network that generates high-frequency groundwater data. However, the operational accessibility of this information remains limited for many stakeholders.

Different users require different forms of groundwater information. Citizens and farmers need simplified local status indicators, district officers require anomaly monitoring and alert systems, while researchers need analytical and comparative tools. Existing platforms such as India-WRIS and IN-GRES remain largely expert-oriented and do not sufficiently support role-specific operational workflows.

This paper presents AQUIS (Aquifer Query and User Information System), a prototype groundwater monitoring and information platform designed to improve accessibility of CGWB DWLR data for citizens, officers, and researchers. The study covers the comparative platform review, system architecture, dataset analysis, interface design process, and heuristic evaluation of the prototype.

The remainder of the paper presents background literature, comparative review of groundwater monitoring systems, design methodology, system architecture, data analysis findings, interface design, usability evaluation, operational implications, and conclusions.

2. Background and Related Work

2.1 Groundwater Monitoring in India

India's groundwater monitoring system is managed by the Central Ground Water Board (CGWB) under the Ministry of Jal Shakti. The 2023 groundwater assessment estimated annual replenishable groundwater resources at approximately 449 billion cubic metres, with annual extraction exceeding 240 billion cubic metres, indicating high utilisation across several regions (CGWB, 2023).

Groundwater extraction in India is driven largely by agricultural demand and decentralised private use, particularly in the Indo-Gangetic plains (Shah et al., 2003). Although CGWB assessment methods are technically robust, their outputs remain difficult to operationalise at district level due to technical complexity and limited accessibility for non-expert users (Kulkarni et al., 2015). Platforms such as India-WRIS have improved national-level data integration, but simplified interfaces and operational monitoring workflows for field-level stakeholders remain limited.

2.2 Design and Development Approach

AQUIS is developed using a structured Design Science Research (DSR) process following Peffers et al. (2007), which provides a practical framework for developing and evaluating software systems that address real-world problems. DSR is used here as a development discipline to ensure the AQUIS prototype is grounded in both domain evidence and stakeholder needs, rather than as a theoretical contribution in itself.

The user interface design follows User-Centred Design (UCD) principles as described in ISO 9241-210:2019. Interface decisions draw on Nielsen's (1994) usability heuristics for practical design review and Shneiderman's (1996) visual information-seeking framework, which structures the map interface across user roles. These are used as practical design tools.

2.3 HCI in Environmental and Civic Platforms

Colour-coded status indicators have been found to improve comprehension among low-literacy users in Indian civic contexts (Medhi et al., 2009), informing the AQUIS status badge design. Voice-based interaction supports users with limited English literacy in rural settings, an approach applied in the AQUIS public interface. Mobile alert systems have proven effective in water resource management contexts (Condon et al., 2021), supporting the AQUIS notification module design.

2.4 Identified Gap

No current Indian platform combines near-real-time DWLR data with role-differentiated interfaces accessible to citizens, district officers, and researchers within a single system. India-WRIS and IN-GRES provide structured groundwater data but are oriented toward expert users and periodic reporting. The comparative review in Section 3 confirms this gap. AQUIS is designed to address the operational accessibility dimension that existing platforms do not currently cover.

3. Comparative Review of Groundwater Monitoring Platforms

To establish the design requirements for AQUIS, this section reviews three major groundwater monitoring platforms alongside India's existing national systems. The comparison focuses on operational usability and citizen accessibility rather than technical infrastructure depth.

3.1 USGS Water Data for the Nation — United States

The USGS Water Data for the Nation (WDFN) provides real-time data from a large monitoring network, with useful features including data customisation, the WaterAlert notification service, and public APIs. WaterAlert supports threshold-based notifications on near-real-time updates. The system does not offer simplified views for non-expert users, has no role-based access structure, and does not classify groundwater stress by extraction rates. Its threshold-based alerting model informed the AQUIS alert module design.

3.2 HidroWeb / SNIRH — Brazil

Brazil's National System of Information on Water Resources (SNIRH) provides historical and near-real-time water data via APIs with integrated quality checking. Its limitations include single-language support (Portuguese only), lack of groundwater stress classification, no simplified citizen interface, and no configurable alert system for end users.

3.3 IN-GRES and India-WRIS — India

India's main groundwater platforms are designed for expert users, focusing on district-level data tables, GIS visualisation, and periodic reporting. Key gaps include the absence of near-real-time DWLR data at high frequency, no mobile application, no citizen-oriented interface, and no configurable alert system. AQUIS uses the same CGWB stress classification underpinning India-WRIS assessments but applies it in near-real time rather than as a periodic report, bridging a meaningful operational gap.

3.4 Feature Comparison

Table I provides a structured feature comparison across all systems evaluated.

Feature	USGS WDFN (USA)	HidroWeb / SNIRH (Brazil)	IN-GRES / India-WRIS	AQUIS (Proposed)
Near real-time data (<1 hr)	Yes	Yes	No	Yes
Role-based access control	No	No	No	Yes
Citizen-simplified view	No	No	No	Yes
Officer operational dashboard	Partial	No	No	Yes
Researcher analytical workspace	Yes	Yes	Partial	Yes
Configurable alert rules	Yes	No	No	Yes
Extraction rate / stress classification	No	No	Partial	Yes
Mobile application (citizen)	Partial	Partial	No	Yes
Voice query interface	No	No	No	Yes
Open REST API	Yes	Yes	Partial	Yes
Spatiotemporal playback	No	No	No	Yes
HCI validation documented	Partial	No	No	Yes

Note. *Source: Authors' comparative analysis (2026). Feature assessment based on platform documentation and operational review.*

The comparison confirms a consistent pattern: existing systems provide strong data infrastructure and researcher-level API access but lack role-differentiated usability and citizen-accessible interfaces. No current system — national or international — combines near-real-time DWLR access, role-based interfaces, and a simplified citizen view. AQUIS is designed to address this combination.

4. Design and Development Methodology

4.1 DSR Process Application

AQUIS follows the six-stage Design Science Research (DSR) process proposed by Peffers et al. (2007), including problem identification, objective definition, design and development, demonstration, evaluation, and communication. The framework is applied pragmatically to support structured system development.

Problem identification is based on the Ministry of Jal Shakti problem context and the comparative review in Section 3. Design and development include system architecture, API planning, and iterative prototyping, while evaluation is conducted through heuristic review (Section 8). The prototype represents the demonstration stage of the process.

4.2 User-Centred Design Process

Within the DSR framework, AQUIS follows a User-Centred Design approach based on ISO 9241-210:2019. The process includes stakeholder analysis, system modelling, architecture design, prototype development, and interface refinement.

Three primary user groups are considered: citizens and farmers, district water officers, and groundwater researchers. These roles are derived from CGWB operational workflows and the Ministry of Jal Shakti problem statement.

4.3 Data Analysis Methodology

The analysis uses the CGWB DWLR Dataset (2023), covering 736 district-level assessment units across India. Data preprocessing, stress classification, rainfall correlation analysis, and hotspot identification were conducted using Python-based analytical tools.

The analytical findings informed interface decisions such as hotspot visualisation, extraction-rate prioritisation, and alert-based workflows. These relationships between analysis outputs and interface design are discussed further in Section 6.

5. System Architecture

5.1 Three-Tier Layered Architecture

AQUIS follows a three-tier architecture comprising data processing, application logic, and presentation layers. This separation allows the user interface to be modified based on usability feedback without requiring changes to the backend data infrastructure — an important consideration for a system intended to integrate with existing CGWB and India-WRIS pipelines.

Table II presents the technology stack and its operational rationale.

Layer	Technology Stack	HCI and DSR Relevance
Data Processing (Analytics Engine)	Python, Pandas, NumPy, GeoPandas, Seaborn, Matplotlib	Generates extraction rates, hotspot rankings, and correlation outputs that directly inform role-specific interface elements, forming the analytical core of the artefact.
Application Backend	Node.js, Express.js, PostgreSQL + PostGIS, JWT authentication, REST API	Provides role-based API endpoints that deliver only relevant data to each user type, reducing cognitive load and supporting usability.
Presentation Layer	React.js + Tailwind CSS (Web), React Native (Mobile), Leaflet/Mapbox, Chart.js / D3.js	Implements role-differentiated interfaces for public users, officers, and researchers.

Note. Source: Authors' system design (2026).

International Journal of Scientific Research in Engineering and Management (IJSREM)
Volume: 10 Issue: 05 | May - 2026 SJIF Rating: 8.659 ISSN: 2582-3930

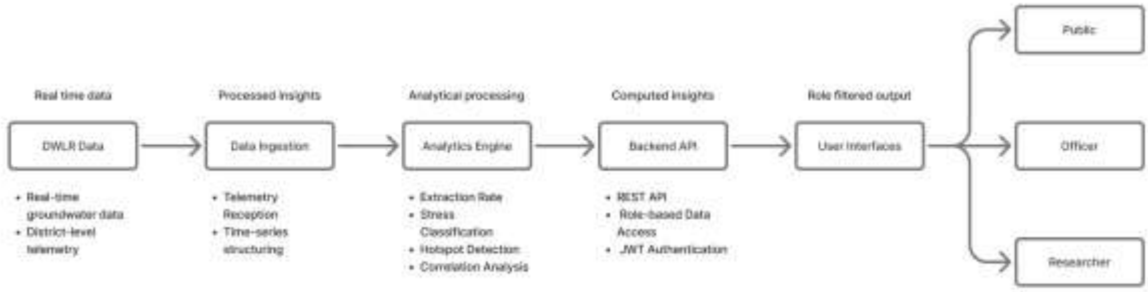

Figure 1. *AQUIS architecture and data flow from DWLR telemetry through analytics, backend services, and role-based interfaces.*

5.2 Module Decomposition

The backend is organised into four functional modules. The Authentication Module enforces role-based session control and determines which interface variant is presented at login. The Ingestion Module validates incoming DWLR telemetry before storage, maintaining data reliability for downstream use.

The Analytics Module provides time-series, correlation, and spatiotemporal data services. The Alert Module delivers proactive notifications via push and email channels, reducing the need for continuous dashboard monitoring and supporting timely awareness for field officers.

5.3 API Design

The backend exposes a versioned REST API secured using JWT-based authentication. Key endpoints include: POST /auth/login; GET /stations/status (role-filtered); GET /analytics/timeseries; GET /analytics/correlation; POST /telemetry/data (secure ingestion); POST /alerts/rules (threshold-based alert configuration). Each endpoint enforces role-based access control, ensuring backend logic mirrors the interface-level differentiation.

6. Data Analysis: From DWLR Data to Design Decisions

A practical feature of the AQUIS development process is the direct connection between data analysis findings and interface design choices. This section documents each analytical result and the corresponding design decision it informed.

6.1 Dataset and Preprocessing

The CGWB DWLR Dataset (2023) includes 736 district-level assessment unit records. Key variables include state, district, rainfall, groundwater availability (fresh and saline), and extraction across domestic, industrial, and irrigation sectors. Preprocessing involved column standardisation, removal of invalid or zero-availability records, and outlier filtering (extraction rates capped at 300% to exclude physically implausible values). The cleaned dataset contains 730 valid records.

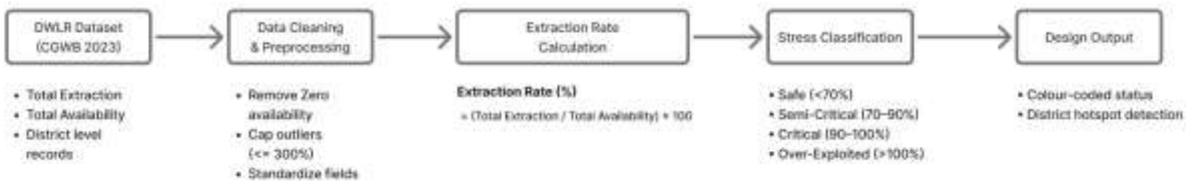

Figure 2. Data processing pipeline from DWLR dataset through preprocessing, extraction rate computation, stress classification, and design outputs. Source: Authors' analysis (2026).

6.2 Extraction Rate and Stress Classification

The primary analytical metric is the groundwater extraction rate: **Extraction Rate (%) = (Total Extraction / Total Availability) × 100.**

Table III reports the distribution of 730 assessment units across CGWB stress categories.

Category	Range	Proportion (%)
Safe	< 70	85.32
Semi-Critical	70–90	5.52
Critical	90–100	1.60
Over-Exploited	> 100	7.56

Note. Source: Authors' analysis of CGWB DWLR Dataset 2023 (n = 730 assessment units after preprocessing).

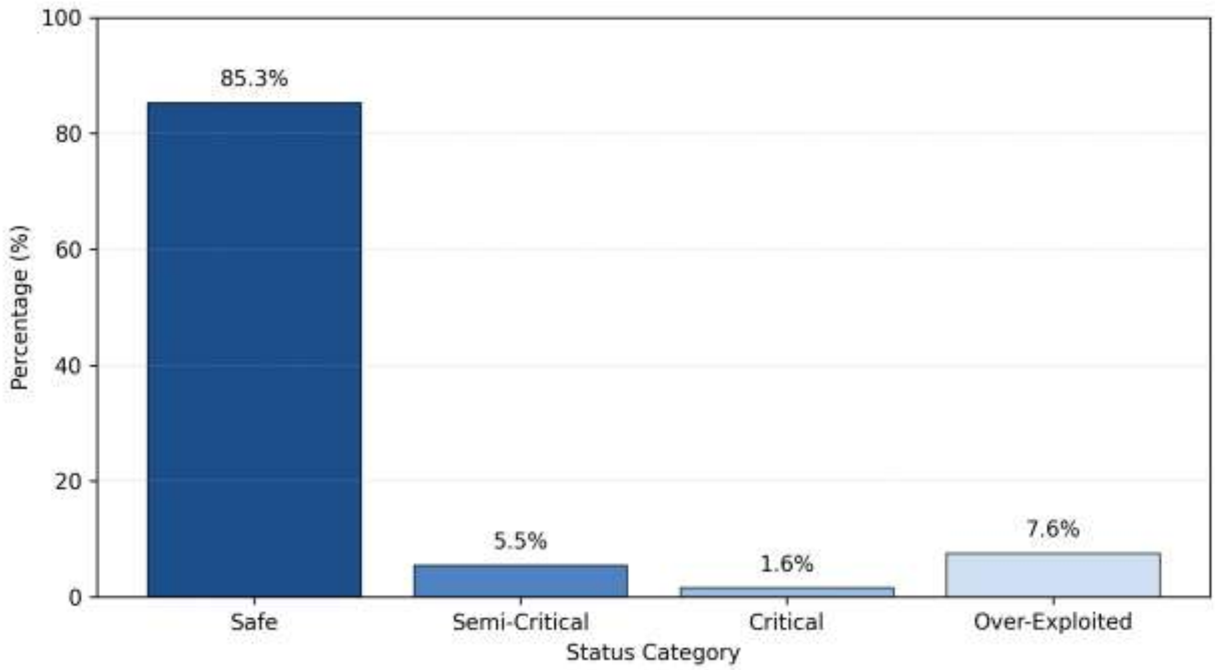

Figure 3. Percentage distribution of groundwater stress categories across 730 assessment units. Source: Authors' analysis of CGWB DWLR Dataset 2023.

Design implication: The four-category classification maps directly to AQUIS's colour-coded status system (Safe = green; Semi-Critical = amber; Critical = orange; Over-Exploited = red). The high proportion of safe units (85.3%) supports a positive default map state, reducing unnecessary alarm while allowing stressed districts to stand out clearly through visual contrast.

6.3 District-Level Hotspot Identification

Assessment units were ranked by mean extraction rate to identify districts with the highest groundwater stress.

Table IV presents the ten most over-exploited districts, which were used to pre-configure default alert thresholds in the officer and researcher interfaces.

Rank	District	State	Mean Extraction Rate (%)
1	West Delhi	Delhi	290.08
2	Arwal	Bihar	274.38
3	Kavaratti	Lakshadweep	273.33
4	North Delhi	Delhi	259.38

5	Munger	Bihar	245.64
6	Ramban	Jammu and Kashmir	233.86
7	Pathanamthitta	Kerala	224.96
8	Wayanad	Kerala	216.32
9	Sirohi	Rajasthan	214.09
10	Mahendragarh	Haryana	212.21

Note. Source: Authors' analysis of CGWB DWLR Dataset 2023.

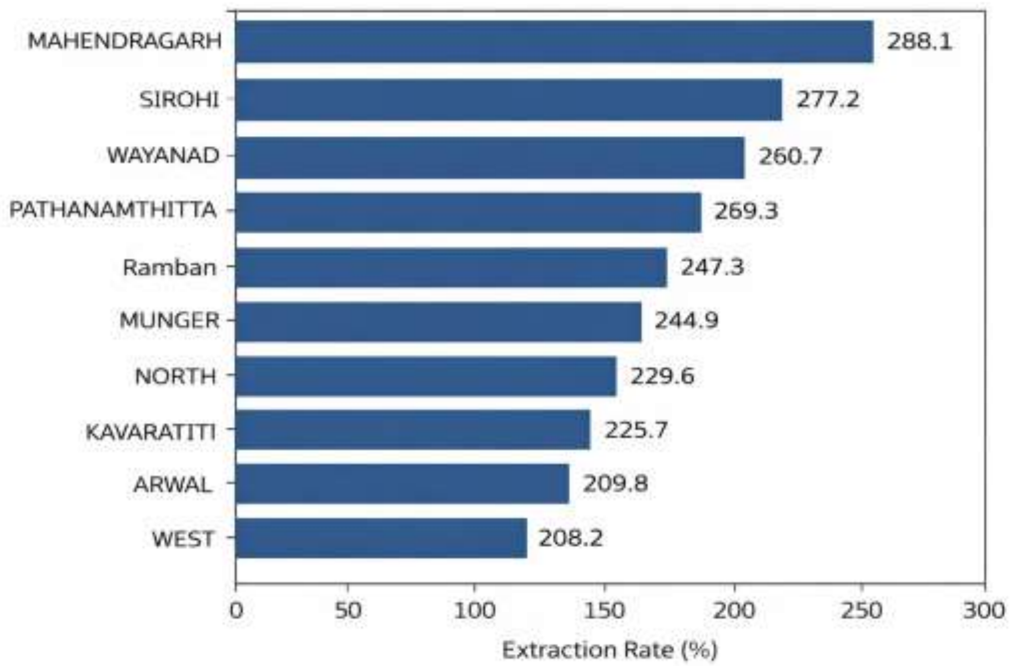

Figure 4. Top 10 over-exploited districts by mean extraction rate (%). Source: Authors' analysis of CGWB DWLR Dataset 2023

Design implication: The hotspot analysis informed the decision to make district-level hotspot ranking a primary feature of officer and researcher dashboards, and to pre-configure alert rules for the top ten districts. The map heat overlay defaults to extraction rate rather than raw water level depth, as extraction rate provides more operationally relevant information for intervention planning. This also reduces alert fatigue by directing attention toward actionable anomalies rather than routine data fluctuations.

6.4 Rainfall–Groundwater Correlation

A Pearson correlation was conducted between district-level total rainfall (mm) and total groundwater availability (hectare-metres) across the 730 cleaned records. The result was **r = 0.003 (p > 0.05) — a negligible correlation.** This finding indicates that, at the district level, groundwater availability is not substantially driven by rainfall. Extraction intensity and land-use patterns appear to be the primary determinants, consistent with prior research on India's groundwater system (Shah et al., 2003; Sharma et al., 2018).

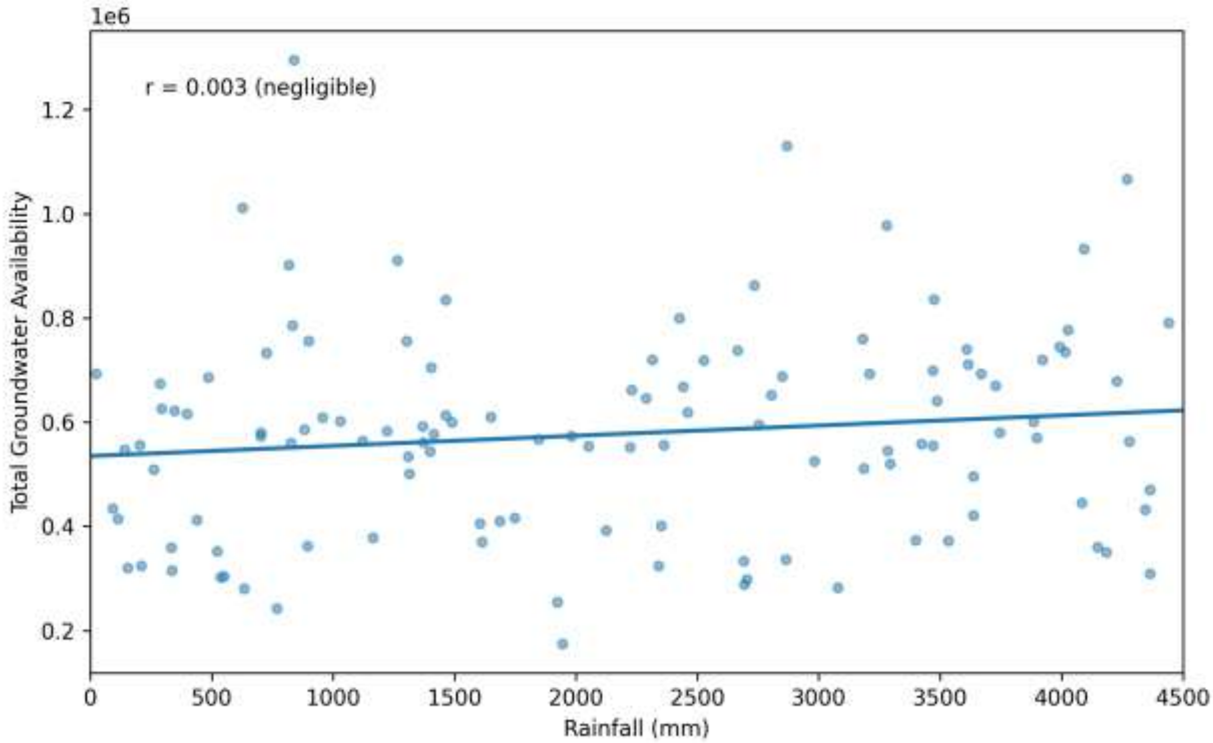

Figure 5. Rainfall vs. total groundwater availability scatter plot — Pearson r = 0.003, near-flat regression line confirming negligible correlation. Source: Authors' analysis of CGWB DWLR Dataset 2023.

Design implication: An initial design assumption placed a rainfall–groundwater correlation view as the primary analytical feature for researchers. The empirical finding of negligible correlation ($r = 0.003$) led to its reclassification as a secondary analytical tool. Extraction-rate-based metrics were prioritised as the primary interface focus, illustrating how data analysis can directly revise interface priorities during the development process.

7. Interface Design and Information Architecture

7.1 Design Principles

The AQUIS interface design applies practical usability principles from Nielsen (1994), Shneiderman (1996), and Norman (2013). These guided the use of colour-coded status indicators, simplified navigation, map-based filtering, and clear system feedback across all user roles.

7.2 Information Architecture by Role

Table V defines the information architecture by user role, showing how interface complexity increases progressively from public users through to researchers while ensuring usability at each level.

Section	Public User	Officer	Researcher
Home / Dashboard	Status badge with trend indicator	Station count, alert feed	Region-wide metrics
Map View	Colour-coded pins	Heat overlay + anomaly detection	Filterable heatmap + timeline
Alerts	Push notifications	Configurable rules	Anomaly annotations
Reports / Analysis	Weekly trends	PDF/CSV export	Correlation workspace
Profile / Settings	Language, voice, location	Station management	API + dataset export

Note. Source: Authors' information architecture design (2026).

7.3 User Flow Design

7.3.1 Onboarding and Authentication

A five-step onboarding sequence collects: (1) language preference (Hindi, English, Marathi, Telugu); (2) basic user information; (3) role selection (General Public, Water District Officer, Researcher); (4) location setup (auto-detect or manual search); and (5) voice assistant opt-in. Only essential information is collected at each step to reduce completion burden. After onboarding, the primary citizen task — checking local groundwater status — requires zero navigation steps from the home screen.

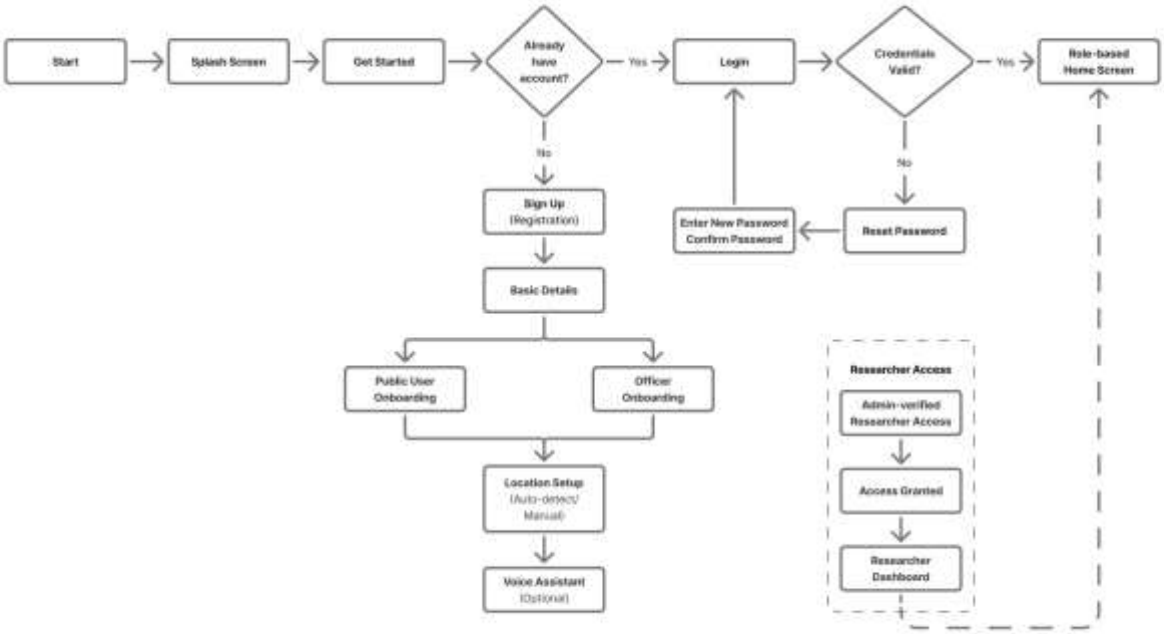

Figure 6. AQUIS onboarding flow from splash screen to role-based home interface, including authentication, role selection, location setup, and voice assistant opt-in.

7.3.2 Public User Flow

The public user mobile flow is deliberately minimal. From the home screen, three primary actions are directly accessible: Map View (station pin interaction to view water level details), Alerts, and Profile/Settings. A voice assistant shortcut supports natural-language queries for users with limited text-based digital literacy, particularly in rural contexts.

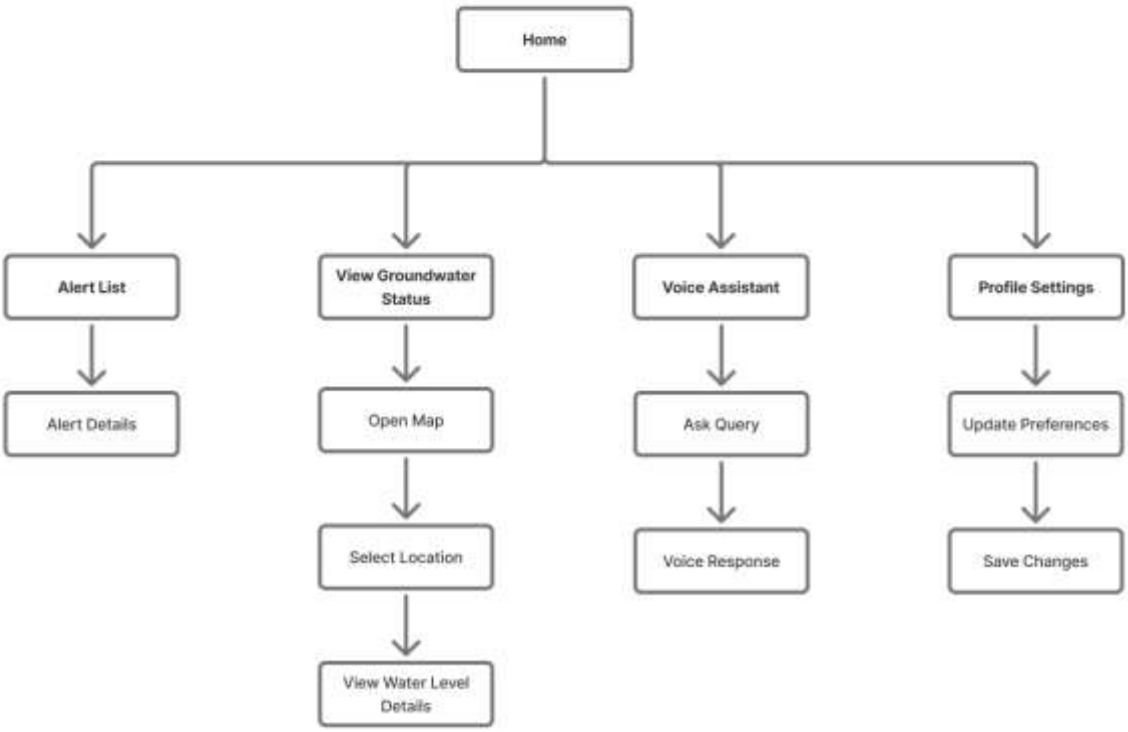

Figure 7. AQUIS Public User interaction flow — home screen with status badge branching to map view, alert screen, voice assistant, and profile settings.

7.3.3 Water District Officer Flow

The officer flow provides greater operational depth. The home dashboard surfaces active critical alerts and a station health summary. Officers access three modules: Map View (with extraction rate heat overlays and anomaly detection), Alert Management (configurable rule definition and alert history), and Reports and Analysis (station comparison, trend visualisation, and PDF/CSV export). Anomaly detection — the officer's primary task — is reachable within two interactions from login.

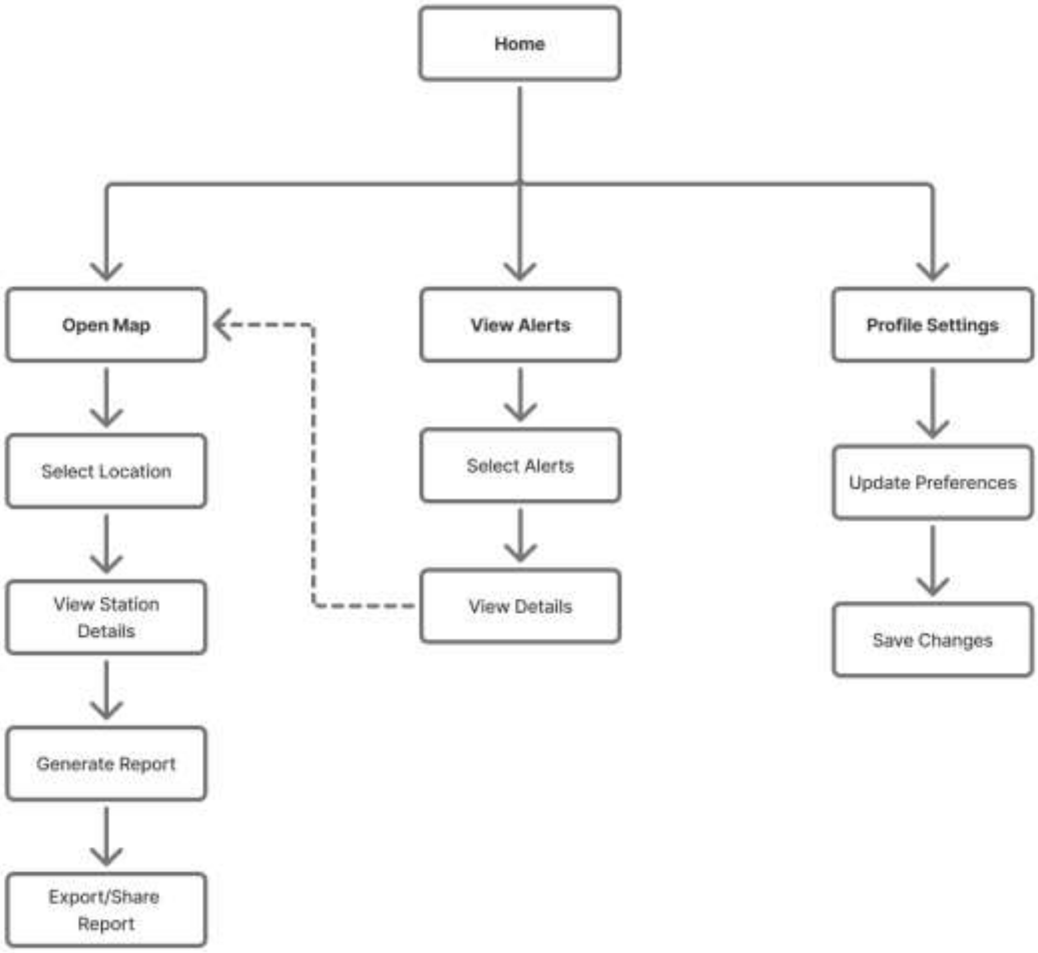

Figure 8. AQUIS Water District Officer interaction flow — home dashboard branching to map view, alert management, reports/analysis, and profile with export and share options.

7.3.4 Researcher Flow

The researcher workspace centres on an Analysis Mode selector from which six analytical workflows branch: time-series analysis, map-based analysis, comparative analysis, insight generation, report generation, and data exploration. Outputs feed into a shared export layer, allowing researcher-derived insights to be packaged as reports for policymakers and officers. A persistent home shortcut is available across all analytical views to prevent disorientation within complex navigation paths.

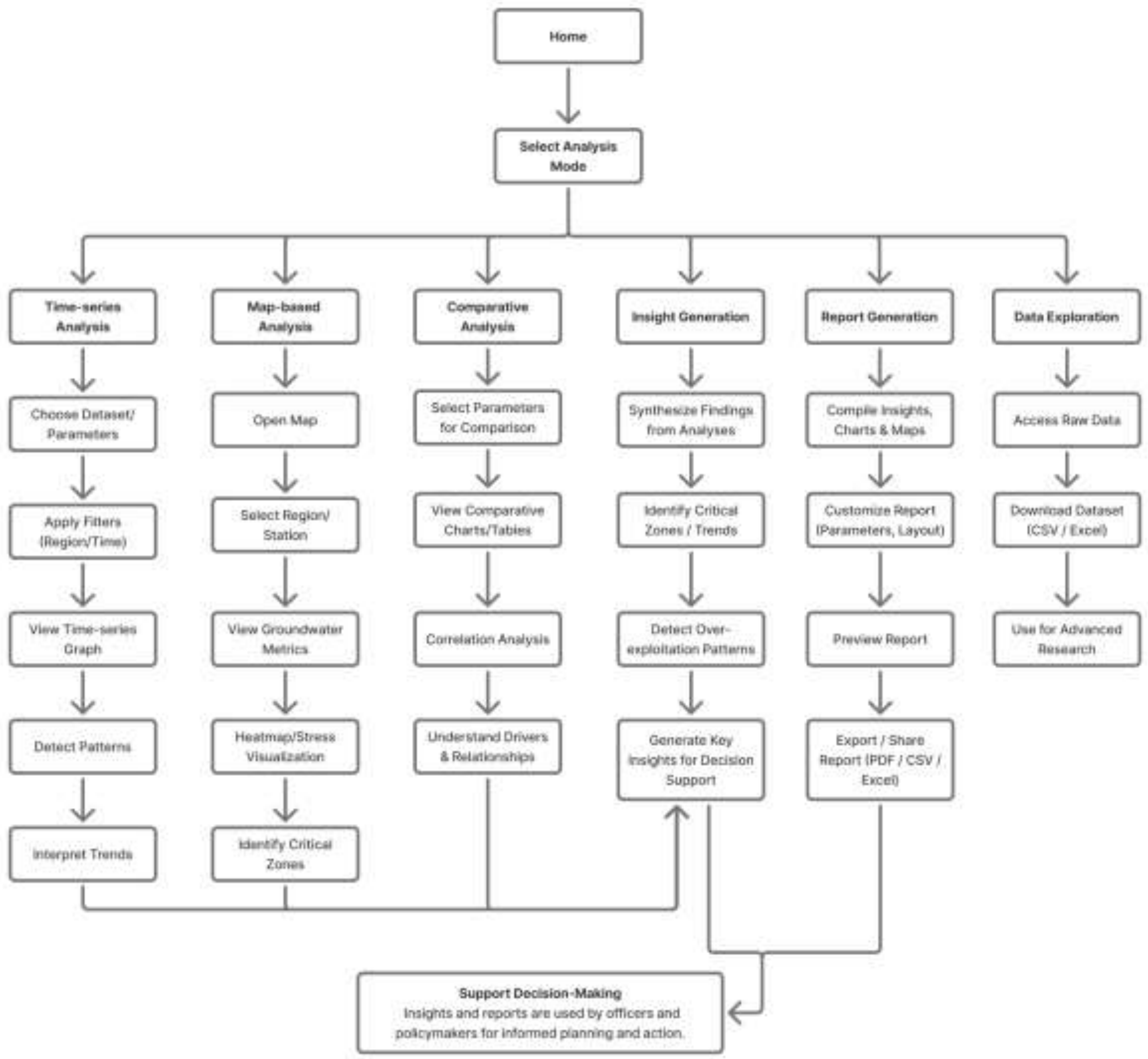

Figure 9. AQUIS Researcher interaction flow — home dashboard branching to map view with playback, analysis workspace (correlation, time-series), voice assistant, alert screen, and profile.

7.4 Wireframe and Prototype Development

Low-fidelity wireframes were developed for both mobile and web interfaces to validate information layout and navigation logic before high-fidelity design work began. The wireframing stage identified a key issue: the initial hamburger menu design required three interactions to reach the alert screen, exceeding the two-interaction target for officer primary tasks. This led to a bottom tab bar redesign with a persistent alert badge — one of six improvements identified during the review cycle.

Figure 10. Low-fidelity wireframes for AQUIS mobile interface, illustrating role-based layout structure for public and officer users.

7.5 High-Fidelity Prototype

7.5.1 Public User Interface

The public mobile interface focuses on simplified groundwater status access. The home screen displays groundwater category, current water level, weekly trend indicators, and nearby DWLR stations with minimal navigation complexity. Map, alert, and voice assistant features are accessible through the bottom navigation bar.

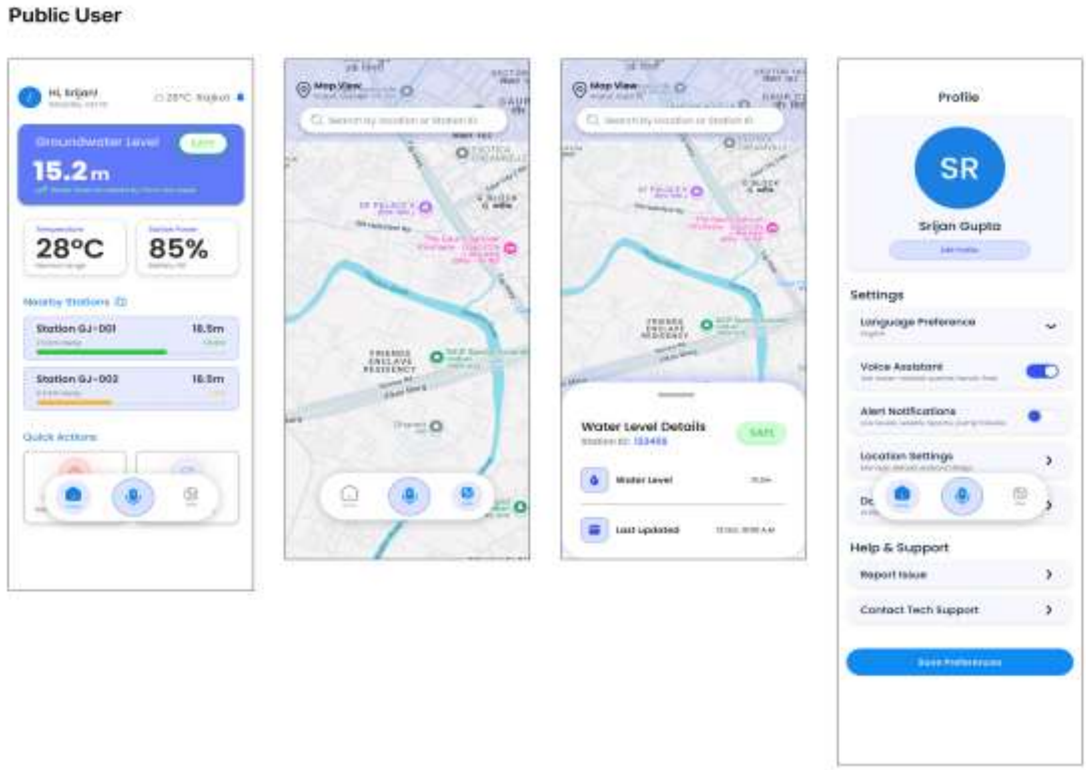

Figure 11. AQUIS public user mobile interface showing key screens: (a) home, (b) map, (c) water level detail, (d) profile/settings.

7.5.2 Water District Officer Interface

The officer dashboard supports operational monitoring through extraction-rate alerts, station summaries, trend visualisation, and map-based hotspot identification. Reporting tools include time-range filters and anomaly annotations to support field-level monitoring and response workflows.

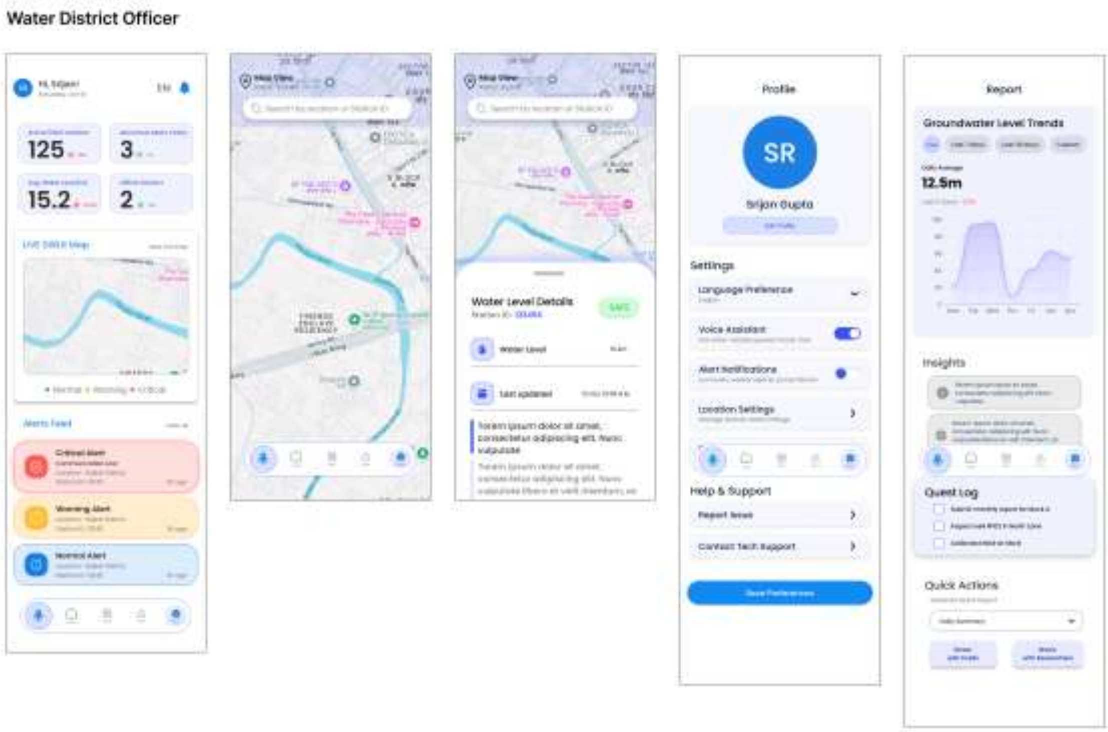

Figure 12. AQUIS officer dashboard interface: (a) home dashboard, (b) map view, (c) water level detail, (d) reporting interface.

7.5.3 Researcher Web Dashboard

The researcher dashboard supports spatial and temporal exploration of DWLR data through map overlays, trend analysis, and correlation tools. A dedicated insights panel highlights detected anomalies and supports analytical reporting workflows.

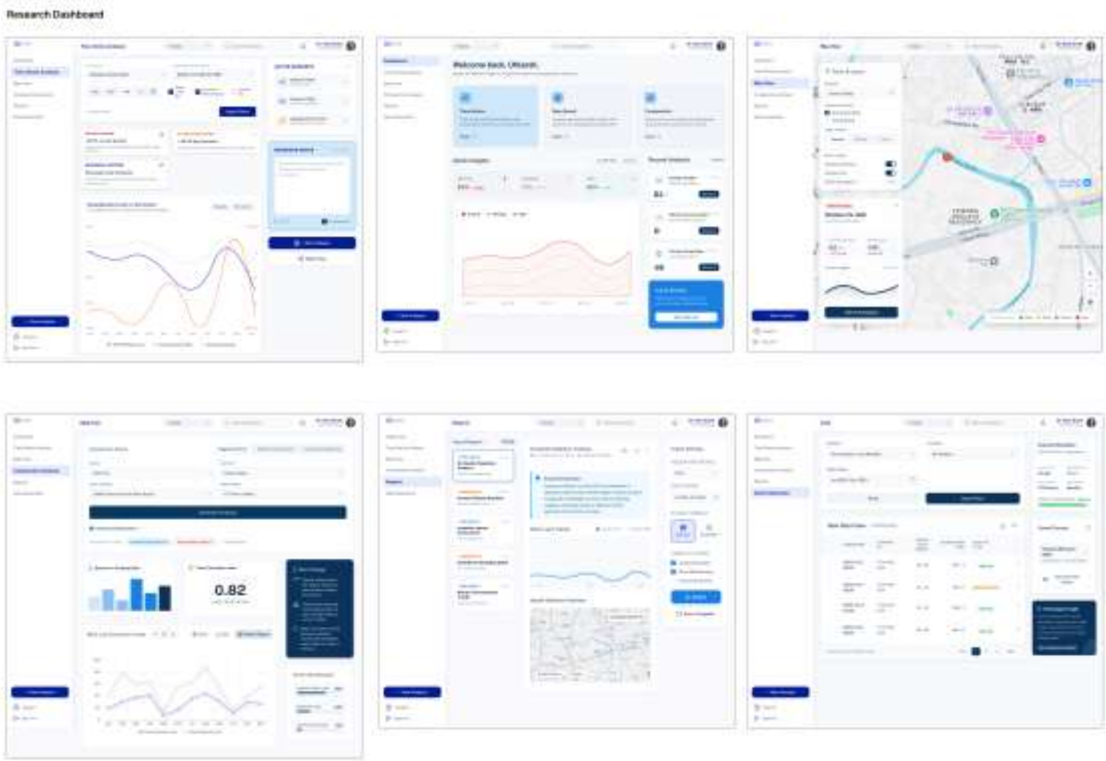

Figure 13. AQUIS Researcher high-fidelity web dashboard — (a) main dashboard with map and trend chart, (b) full-screen map with filter controls, (c) correlation and insights panel.

8. Usability Evaluation

The AQUIS prototype was evaluated through an expert heuristic review based on Nielsen's (1994) usability principles. The review identified six usability issues and corresponding interface refinements, summarised in Table VI.

Heuristic	Issue Identified	Design Change Made
H1: Visibility of System Status	No latency indicator	Added "Last updated" timestamp
H2: Match with Real World	"HAM" unclear	Changed to "hectare-metres (ham)" with tooltip
H4: Consistency	Different colour schemes	Harmonised colour system
H5: Error Prevention	Invalid thresholds allowed	Added validation constraints
H8: Minimalist Design	Too many elements	Reduced to 4 primary elements
H10: Help & Documentation	No help section	Added Help + voice assistant

Note. Source: Authors' heuristic evaluation (2026). Heuristic definitions from Nielsen (1994).

Two additional observations emerged during evaluation. The officer dashboard initially prioritised raw water depth instead of extraction rate, which was revised to better support hotspot identification. The researcher correlation module also lacked contextual explanation of statistical significance and was updated with tooltip guidance.

The current study focused on system architecture, workflow modelling, and prototype refinement prior to deployment-stage testing. Future work will include structured usability testing with representative stakeholders and live DWLR telemetry integration.

9. Discussion

9.1 AQUIS Within the Indian Groundwater Monitoring Landscape

The comparative review indicates that existing platforms such as India-WRIS and IN-GRES remain primarily expert-oriented and periodic in their operational use. AQUIS is designed to extend these systems through role-specific interfaces and simplified access to groundwater monitoring information for citizens, officers, and researchers.

India's DWLR network already generates large volumes of monitoring data; however, operational accessibility remains limited for many stakeholders. AQUIS addresses this challenge by translating groundwater telemetry and assessment outputs into more usable monitoring workflows without requiring changes to existing CGWB or India-WRIS infrastructure.

9.2 Data-to-Design Mapping as a Practical Workflow

Table VII maps the principal analytical findings from Section 6 to their corresponding design outcomes. This data-to-design traceability is useful for a monitoring system where interface priorities should reflect ground conditions rather than design assumptions alone.

Analytical Finding	Design Decision	Rationale
85.3% Safe; 7.6% Over-Exploited	Default positive map + red hotspots	Focus on actionable regions
Rainfall correlation negligible (r = 0.003)	Extraction prioritised	Avoid misleading weak predictors
High extraction districts	Hotspot ranking + alerts	Prioritise high-risk regions
Urban over-exploitation	District-level context added	Improve comparative understanding

Note. Source: Authors' data-to-design mapping (2026).

This approach is consistent with the recommendation by Kulkarni et al. (2015) that CGWB assessment outputs need to be translated into formats that are operationally useful at the district level. The data-to-design mapping provides one practical model for how this translation can be structured during system development.

9.3 Operational Implications for Groundwater Management

AQUIS provides role-specific interfaces for officers, citizens, and researchers within a shared groundwater monitoring framework. For district officers, the platform supports extraction-rate alerts, hotspot identification, and trend monitoring to assist field-level response. The citizen interface focuses on simplified local groundwater status access with minimal navigation and optional voice assistance for users with limited digital literacy.

For researchers, the analytical workspace supports spatial and temporal exploration of DWLR data along with report generation for operational use. At the infrastructure level, AQUIS is designed as an interface layer compatible with existing CGWB and India-WRIS systems through standard REST APIs, reducing integration complexity and supporting interoperability within Digital India frameworks.

9.4 Implications for Indian Water Governance

AQUIS aligns with the National Water Policy (2012) emphasis on transparency and community participation in water resource management. By providing a citizen-accessible interface to DWLR data, the platform supports greater public awareness of local groundwater conditions, which may contribute to more informed community-level responses to groundwater stress.

At the governance level, the role-based architecture reflects the multi-stakeholder structure of Indian groundwater administration, where citizens, field officers, district administrators, and researchers operate with different information needs and analytical capacities. A platform that supports all three groups within a shared data infrastructure may reduce the gap between monitoring data and operational decision-making that has been noted in studies of Indian groundwater governance (Kulkarni et al., 2015).

9.5 Limitations

The empirical analysis uses batch-processed historical data rather than live DWLR telemetry. Real-time integration with CGWB telemetry infrastructure is an important next step and is identified as future work. The usability evaluation relied on expert heuristic review; structured testing with district officers, researchers, and citizen users is needed before deployment-phase conclusions can be drawn. District-level aggregation in the DWLR dataset may also obscure sub-district variability relevant to hydrological analysis in certain contexts. The current mobile prototype is designed for Android; cross-platform development for iOS and web mobile is also future work.

10. Conclusion

AQUIS highlights that the challenge in Indian groundwater monitoring is increasingly one of accessibility rather than data availability. While the DWLR network generates substantial monitoring data, its operational use remains limited across different stakeholder groups.

The proposed platform combines role-specific interfaces, groundwater stress visualisation, hotspot mapping, and configurable alerts within a unified monitoring framework compatible with existing CGWB and India-WRIS systems. The study also illustrates how groundwater assessment outputs can support practical interface development for environmental monitoring applications.

Future work will focus on live DWLR telemetry integration, deployment-stage usability testing, and predictive analytics for early-warning support within groundwater management workflows.

Acknowledgements

AI-assisted language editing tools were used for editorial refinement. All analysis, interpretation, and conclusions remain the responsibility of the authors.

REFERENCES

Agência Nacional de Águas e Saneamento Básico (ANA) (2026), Sistema Nacional de Informações sobre Recursos Hídricos (SNIRH) — HidroWeb, ANA, Brasília, available at: https://www.snirh.gov.br/hidroweb (accessed April 2026).

Bangor, A., Kortum, P.T. and Miller, J.T. (2008), "An empirical evaluation of the system usability scale", International Journal of Human-Computer Interaction, Vol. 24 No. 6, pp. 574–594.

Central Ground Water Board (CGWB) (2023), Dynamic Ground Water Resources of India 2023, Ministry of Jal Shakti, Government of India, New Delhi, available at: https://cgwb.gov.in (accessed April 2026).

Condon, L.E., Kollet, S., Bierkens, M.F.P., Fogg, G.E., Maxwell, R.M., Hill, M.C., Fransen, H.J.H., Verhoef, A., Van Loon, A.F., Sulis, M. and Abesser, C. (2021), "Global groundwater modeling and monitoring: opportunities and challenges", Water Resources Research, Vol. 57 No. 12, p. e2020WR029500.

Hevner, A.R., March, S.T., Park, J. and Ram, S. (2004), "Design science in information systems research", MIS Quarterly, Vol. 28 No. 1, pp. 75–105.

ISO (2019), Ergonomics of Human-System Interaction — Part 210: Human-Centred Design for Interactive Systems (ISO 9241-210:2019), International Organisation for Standardisation, Geneva.

Kulkarni, H., Shah, M. and Vijay Shankar, P.S. (2015), "Shaping the contours of groundwater governance in India", Journal of Hydrology: Regional Studies, Vol. 4, pp. 172–192.

Medhi, I., Prasad, A. and Toyama, K. (2009), "Optimal audio-visual representations for illiterate users of computers", Proceedings of the 18th International Conference on World Wide Web, ACM, New York, NY, pp. 873–882.

Ministry of Electronics and Information Technology (MeitY) (2015), Digital India Programme: Policy Framework and Architecture, Government of India, New Delhi, available at: https://digitalindia.gov.in (accessed April 2026).

Ministry of Water Resources, Government of India (2012), National Water Policy 2012, Government of India, New Delhi, available at: https://jalshakti-dowr.gov.in (accessed April 2026).

National Water Informatics Centre (NWIC) (2024), India-WRIS: Water Resources Information System of India, Ministry of Jal Shakti, Government of India, New Delhi, available at: https://indiawris.gov.in (accessed April 2026).

Nielsen, J. (1994), Usability Engineering, Academic Press, Boston, MA.

Norman, D.A. (2013), The Design of Everyday Things, revised and expanded ed., Basic Books, New York, NY.

Peffers, K., Tuunanen, T., Rothenberger, M.A. and Chatterjee, S. (2007), "A design science research methodology for information systems research", Journal of Management Information Systems, Vol. 24 No. 3, pp. 45–77.

Shah, T., Roy, A.D., Qureshi, A.S. and Wang, J. (2003), "Sustaining Asia's groundwater boom: an overview of issues and evidence", Natural Resources Forum, Vol. 27 No. 2, pp. 130–141.

Sharma, B.R., Gulati, A., Mohan, G., Manchanda, S., Ray, I. and Amarasinghe, U. (2018), Water Productivity Mapping of Major Indian Crops, NABARD-ICRISAT Report, National Bank for Agriculture and Rural Development, Mumbai.

Shneiderman, B. (1996), "The eyes have it: a task by data type taxonomy for information visualizations", Proceedings of the IEEE Symposium on Visual Languages, pp. 336–343.

US Geological Survey (USGS) (2026), Water Data for the Nation (WDFN), US Geological Survey, Reston, VA, available at: https://waterdata.usgs.gov (accessed April 2026).